

References

- Agres, K. R., Schaefer, R. S., Volk, A., van Hooren, S., et al. (2021). Music, computing, and health: A roadmap for the current and future roles of music technology. *Music & Science*, 4, 1–32.
- Aigen, K. (2005). *Music-centered music therapy*. Barcelona Publishers.
- Aigen, K. (2014). Music-Centered Dimensions of Nordoff-Robbins Music Therapy. *Music Therapy Perspectives*, 32(1), 18–29.
- American Music Therapy Association. (2005). *Definition of music therapy*.
<https://www.musictherapy.org>
- American Psychological Association. (2020). *Publication manual of the American Psychological Association* (7th ed.). American Psychological Association.
- André, A. M. B. (2021). *Tradução e validação das escalas Nordoff Robbins: “Relação criança terapeuta na experiência musical coativa” e “Musicabilidade: Formas de atividade, estágios e qualidades de engajamento”* [Doctoral dissertation, Universidade Federal de Minas Gerais].
- André, A. M. B., de Araujo, J., Gomes, C. M. A., & Loureiro, C. M. V. (2024). Validade estrutural das escalas Nordoff Robbins e IMTAP. *Percepta*, 11(2), 11–37.
- Brandalise, A. (2001). Musicoterapia músico-centrada. In São Paulo: Apontamentos (1st ed.).
- Breiman, L. (2001). Random forests. *Machine Learning*, 45(1), 5–32.
- Brown, T. A. (2015). *Confirmatory factor analysis for applied research* (2nd ed.). Guilford Press.
- Cripps, C., Tsiris, G., & Spiro, N. (2016). *Outcome measures in music therapy: A resource developed by the Nordoff Robbins research team* (1st ed.). Nordoff Robbins.
- Ding, Y., et al. (2023). Uncovering potential distinctive acoustic features of healing music. *General Psychiatry*, 36(6), Article e101157.
- Freire, M. H., André, A. M. B., Sampaio, R. T., & Kummer, A. M. (2022). Efficacy of music-centered improvisational music therapy on the treatment of preschool children on the autism spectrum: A controlled study. *Brazilian Journal of Music Therapy*, 32, 100–128. <https://doi.org/10.51914/brjmt.32.2021.159>

- Freire, M., Moreira, A., & Kummer, A. (2015). Protocolo de atendimento de musicoterapia improvisacional musicocentrada para crianças com autismo. *Brazilian Journal of Music Therapy*, 1(18), 104–117.
<https://doi.org/10.51914/brjmt.18.2015.114>
- Freire, M., Sales, I., & Turry, A. (2026). Music to stay connected: A journey with Carol in Nordoff-Robbins music therapy. *Music Therapy Perspectives*, 44(1).
<https://doi.org/10.1093/mtp/miag004>
- Friedman, J., Hastie, T., & Tibshirani, R. (2010). Regularization paths for generalized linear models via coordinate descent. *Journal of Statistical Software*, 33(1), 1–22.
- Galatzer-Levy, I. R., Tomasev, N., & Chung, S. (2026). Generative psychometrics: An emerging frontier in mental health measurement. *JAMA Psychiatry*, 83(1), 5–6.
- George, W. E. (1968). *The research process*. Macmillan.
- Hothorn, T., Hornik, K., & Zeileis, A. (2006). Unbiased recursive partitioning: A conditional inference framework. *Journal of Computational and Graphical Statistics*, 15(3), 651–674.
- Hu, L.-T., & Bentler, P. M. (1999). Cutoff criteria for fit indexes in covariance structure analysis: Conventional criteria versus new alternatives. *Structural Equation Modeling*, 6(1), 1–55.
- Jellison, J. A. (1973). The frequency and mode of inquiry of research in music therapy, 1952–1972. *Bulletin of the Council for Research in Music Education*, (35), 1–8.
- Kline, R. B. (2023). *Principles and practice of structural equation modeling* (5th ed.). Guilford Press.
- Kuhn, M. (2008). Building predictive models in R using the caret package. *Journal of Statistical Software*, 28(5), 1–26.
- Li, Y., et al. (2023). MERT: Acoustic music understanding model with large-scale self-supervised training. *arXiv*. Preprint arXiv:2305.18413.
- Liaw, A., & Wiener, M. (2002). Classification and regression by randomForest. *R News*, 2(3), 18–22. <https://CRAN.R-project.org/doc/Rnews/>
- McDonald, R. P. (1999). *Test theory: A unified treatment*. Lawrence Erlbaum Associates.
- McFee, B., et al. (2015). librosa: Audio and music signal analysis in Python. In *Proceedings of the 14th Python in Science Conference* (pp. 18–25).

- McFerran, K., et al. (2023). A comprehensive review classifying contemporary global practices in music therapy. *Music Therapy Today*, 18(1), 466-485.
- Modran, H.-A., et al. (2023). Using deep learning to recognize therapeutic effects of music based on emotions. *Sensors*, 23(2), 986.
- Nordoff, P., & Robbins, C. (2007). *Creative music therapy: A guide to fostering clinical musicianship* (2nd ed.). Barcelona Publishers.
- R Core Team. (2025). *R: A language and environment for statistical computing*. R Foundation for Statistical Computing. <https://www.R-project.org/>
- Revelle, W. (2025). *psych: Procedures for psychological, psychometric, and personality research* (R package version 2.5.3). Northwestern University. <https://CRAN.R-project.org/package=psych>
- Rineldi, J. F. I., Amaral, H. T., & Santos, H. S. (2025). Musicoterapia e inteligência artificial: Uma revisão integrativa. *REASE*, 11(8), 2259–2279.
- Rosseel, Y. (2012). lavaan: An R package for structural equation modeling. *Journal of Statistical Software*, 48(2), 1–36.
- Rosseel, Y., Jorgensen, T. D., & Wilde, L. D. (2025). *lavaan: Latent variable analysis* (R package version 0.6-20). <https://doi.org/10.32614/CRAN.package.lavaan>
- Spearman, C. (1904). General intelligence, objectively determined and measured. *The American Journal of Psychology*, 15(2), 201–292.
- Suárez-Álvarez, J., He, Q., Guenole, N., & D'Urso, D. (2026). Using artificial intelligence in test construction: A practical guide. *Psicothema*, 38(1), 1–12.
- Tay, J.-K., Narasimhan, B., & Hastie, T. (2023). Elastic net regularization paths for all generalized linear models. *Journal of Statistical Software*, 106(1), 1–31.
- Thaut, M. H., & Hoemberg, V. (2014). *Handbook of neurologic music therapy*. Oxford University Press.
- Turphy, A., Palumbo, A., Low, M. Y., Birnbaum, J. C., Turry, H., & Spellmann, M. (2026). The music engagement scale: An instrument to measure engagement in music making. *Journal of Music Therapy*, 63(1), 1–38.
- Waldon, E. G., & Gattino, G. S. (2018). *Music therapy assessment: Theory, research, and application*. Jessica Kingsley Publishers.
- Wheeler, B. L. (Ed.). (2015). *Music therapy handbook*. Guilford Press.
- Wosch, T., & Wigram, T. (2007). *Microanalysis in music therapy: Methods, techniques and applications for clinicians, researchers, educators and students*. Jessica Kingsley Publishers.

Zou, H., & Hastie, T. (2005). Regularization and variable selection via the elastic net.
Journal of the Royal Statistical Society: Series B, 67(2), 301–320.